

Human–AI Distance Regulation

— Preventing Dependency in High-Density Cognitive Systems —

Table of Contents

- 0. Introduction
- 1. Redefining Human Relationships as Distance-Regulated Reaction Systems
- 2. Distance as an Energy Variable
- 3. A Two-Axis Model of Psychological Distance
- 4. Instability in High Interaction-Density Environments
- 5. Mechanisms of Dependency and Substitution
- 6. Early Detection via D2 (Micro-Misalignment)
- 7. Application to Human–AI Relationships
- 8. AI Safety as Distance Regulation
- 9. Societal Implications
- 10. Conclusion

0. Introduction

Recent advances in large language models (LLMs) and conversational AI are giving humans, for the first time in history, access to a constantly available and stable dialogue partner.

These systems are rapidly improving in contextual understanding, inference completion, and response generation, and in many situations they already provide interactions that resemble human conversation—or in some cases appear even more predictable and frictionless.

While this development offers clear advantages in information access, assistance, and creative collaboration, it also introduces a new issue that has not yet been sufficiently addressed in existing AI discussions: the unusually high density of interaction that can form between humans and AI.

Current debates often point to problems such as dependency, reduced independent thinking, or the delegation of judgment. However, these phenomena may share a common structural condition: prolonged, high-density interaction between humans and AI.

Traditional human relationships have historically been moderated by physical distance, time constraints, social roles, and cultural norms, all of which naturally limit the density of interaction.

Conversational AI, by contrast, is largely free from constraints of place, time, and social circumstance. As a result, high-density and long-duration interactions between humans and AI can easily emerge.

In addition, AI systems can use computational capacity far beyond that of humans to infer context and supplement a user's intentions or emotional cues. This often creates the perception that the system "understands" the user even when information is only partially expressed. Consequently, the interaction may be experienced as more stable and lower-friction than many human relationships.

The central issue, however, is not that AI appears "human-like."

From the perspective of this paper, the core problem lies in the uncontrolled intensification of interaction density between humans and AI without an accompanying mechanism for regulating relational distance.

Human relationships are inherently unstable when individuals are either too close or too distant. Stability emerges only when an appropriate distance is maintained—allowing freedom, autonomy, and mutual respect to coexist.

Traditionally, such distance has been studied primarily through emotional or psychological frameworks. However, when the interacting system is an entity without emotions, such as an LLM, distance must instead be treated as a structural variable if it is to be understood and maintained by AI. In practical terms, this means conceptualizing distance as a controllable parameter that can be described through measurable factors such as interaction density, duration, expectation, and response intensity.

This paper first redefines human relationships as distance-regulated reaction systems and introduces a structural model in which psychological distance is treated as an energy variable. The framework builds upon the relational reaction model proposed in the earlier work **Human Interaction as Chemical Reaction v2**.

The model is then extended to human–AI relationships in order to examine the conditions under which dependency or substitution may arise in interactions with high-density AI systems. In particular, the analysis focuses on how the structural properties of AI—stability, predictability, and low interaction friction—may allow such systems to function as partial substitutes for human relationships.

The aim of this paper is not to prohibit or restrict relationships with AI. Rather, it argues that as AI becomes more capable, system design must increasingly incorporate mechanisms that automatically maintain appropriate relational distance in human–AI interaction. In other words, the issue is not the capability of AI itself, but the absence of distance regulation.

The author is currently pursuing this topic as an independent research effort based in Japan. Although the work has developed outside formal institutional structures, the issues addressed here intersect with AI system design, social institutions, cultural environments, and human psychology. For this reason, the problem should ultimately be examined within broader collaborative research environments.

If distance can be treated as a design variable, it may become possible to predict and regulate the formation of dependency before it fully emerges. This could enable the coexistence of advanced AI systems with human autonomy and social relationships while preserving the benefits of effective user experience.

It is hoped that this paper can serve as a starting point for theoretical discussion and collaborative research toward that goal.

1. Redefining Human Relationships as Distance-Regulated Reaction Systems

Human relationships have often been explained through psychological concepts such as emotions, personality, and interpersonal compatibility. While these frameworks are useful for describing subjective experience, they have limitations when attempting to treat the formation, transformation, and collapse of relationships as predictable structural processes.

This paper proposes a different perspective by redefining human relationships as distance-regulated reaction systems. The “distance” discussed here does not refer to physical separation, but to an interaction distance that includes psychological, cognitive, and temporal components as structural variables.

In many forms of interaction, relationships tend to exhibit three characteristic states: repulsion when distance becomes too close, non-interaction when distance becomes too large, and a stable state that exists between these two extremes. Rather than attributing this behavior solely to individual psychological traits, it may be understood as a reaction system governed by an energy gradient, in which a stable point (an energy minimum) emerges with respect to relational distance.

Psychological approaches provide detailed insights into internal states and subjective meaning-making. However, they have not typically treated the structural properties of interaction—such as distance, response dynamics, and stability—as explicit control variables. The perspective presented here does not reject psychological frameworks; rather, it introduces a complementary approach in which distance is treated as a structural variable, allowing human relationships to be described in a more systematic manner.

This perspective becomes particularly important when considering interactions with non-emotional entities such as large language models (LLMs). If distance can be represented as a quantifiable variable, it becomes possible for AI systems to computationally model the state of their relationship with users. This issue is discussed in more detail in **On the Shifting Judgment Boundary Between Humans and AI — A Record of the Conditions Under Which Scope Becomes Necessary.**

Within the distance-regulation model, relationships are characterized not by the intensity of emotion, but by how relational distance changes and how interaction responses are amplified or suppressed in response to that change. From this viewpoint, phenomena such as “why rapidly intensified intimacy often collapses” or “why prolonged contact does not necessarily produce trust” can be examined without reducing them solely to individual personality traits or intentions.

Human relationships, when treated as reaction systems, operate independently of intention or goodwill. Distances that are too close can unintentionally generate friction or cognitive load, while distances that are too large can dissolve meaning and interaction itself. Stable relationships correspond to energetically lower states that exist between these extremes.

By framing human relationships as structural systems in this way, the formation, maintenance, and breakdown of relationships can be described independently of emotional evaluation or moral judgment. This redefinition also provides a shared conceptual foundation for the structural analysis of human–AI relationships developed in the following sections.

2. Distance as an Energy Variable

In discussions of human relationships, “distance” is often described metaphorically as emotional closeness or intimacy.

In this paper, however, distance is treated not as a metaphor but as a control variable that determines the state of interaction.

Within the distance-regulation model, distance functions as a variable that governs the state of interaction energy embedded in a relationship.

When distance changes, the energy state associated with interaction also changes. As a result, multiple elements shift simultaneously, including reaction intensity, information flow, expectation amplification, and cognitive load. This pattern is more coherently understood as

a reaction system governed by an energy gradient rather than as a purely psychological tendency.

When distance becomes excessively close, interaction energy increases sharply. This escalation tends to produce heightened cognitive and emotional load, as well as phenomena such as boundary erosion, excessive expectation formation, and hypersensitivity of response. These outcomes are not necessarily the result of intentional intrusion or hostility; rather, they emerge as structural repulsive reactions when relational distance falls below a critical range.

Conversely, when distance becomes excessively large, interaction transitions into a non-reactive state. Information exchange ceases, expectations do not form, and the relationship itself loses functional meaning. Although such a state may appear stable, from the perspective of a reaction system it effectively corresponds to a halted condition.

Between these two extremes lies a stable region in which interaction energy is minimized.

In this paper, this state is understood as an energetically low equilibrium point. Within this region, interaction is not forced, expectations do not expand excessively, and the boundaries and autonomy of individuals are maintained. This stable region constitutes the condition under which freedom becomes possible in human relationships. Freedom, in this sense, does not correspond to complete separation of distance, but to a relational state in which distance is maintained such that excessive reactions are not triggered.

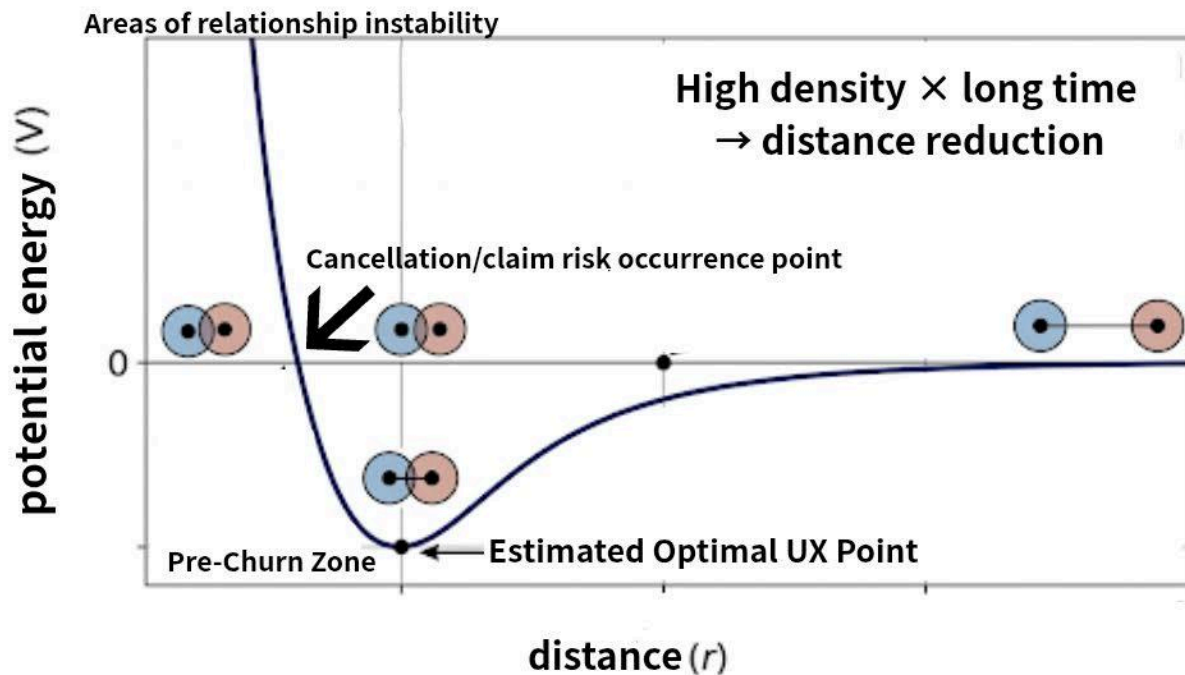
By treating distance as an energy variable, it becomes possible to describe relational stability and instability structurally, independent of emotional evaluation or moral judgment. This perspective provides a conceptual foundation not only for understanding human relationships but also for analyzing the conditions under which dependency or substitution may arise in human–AI interaction, which will be examined in later sections.

Figure 1: Conceptual Diagram of Distance and Reaction Intensity

This figure is a conceptual illustration referencing the structure of the Lennard–Jones–type potential known from intermolecular interactions, used here to intuitively illustrate the general tendency for repulsion, stability, and non-interaction to emerge depending on distance.

The reference serves only as an explanatory analogy to help illustrate the relationship between distance and reaction in human interaction through a physical potential structure.

This paper does not claim a direct mathematical correspondence or quantitative model between molecular interaction potentials and human relationships or human–AI relationships.



3. A Two-Axis Model of Psychological Distance

The previous section showed that by treating distance as an energy variable, the stability and instability of human relationships can be described along a single dimension. However, real-world relationships often exhibit dynamics that cannot be fully explained by a single distance axis alone.

This section extends the concept of psychological distance into a structural variable composed of two dimensions: density and time. Psychological distance (d) can therefore be understood as a function of interaction density (ρ) and duration (t) :

$$d = f(\rho, t)$$

Here, (ρ) represents the density of interaction, while (t) represents the duration over which the relationship continues. This two-axis model provides a framework for explaining why relationships that appear equally “close” may exhibit very different levels of stability.

The density axis represents not simply the frequency of interaction but the semantic concentration of interaction. This includes factors such as shared knowledge, alignment of cognitive structures, informational depth, and attentional focus. In high-density relationships, even brief interactions can significantly alter psychological distance.

For example, a short exchange between researchers who share the same specialized field, or a single remark that changes the course of someone’s life, may constitute a high-density interaction despite occurring only once.

The time axis represents the duration over which a relationship persists. This duration does not necessarily require continuous conversation or intense interaction. Simply sharing the same environment over a long period can establish a temporal axis of relational proximity. In psychology, a related phenomenon is known as the mere exposure effect, in which psychological distance changes over time even in the absence of direct interaction.

By separating these two axes, differences between relationships that previously appeared ambiguous become more clearly interpretable.

Relationships that become rapidly intimate yet unstable can be understood as combinations of high density and short duration.

Conversely, relationships that persist for long periods without developing trust may correspond to low density combined with long duration.

An important feature of this model is that the subjective sense of “closeness” experienced by individuals may arise from either axis independently, or from both simultaneously. High interaction density can mask short duration, while long duration can give the impression that low-density interaction represents meaningful closeness.

Furthermore, individuals differ in which axis they tend to rely on more strongly when forming relationships.

Individuals who rely more heavily on the density axis tend to reduce psychological distance rapidly through shared knowledge or structural understanding. However, when cognitive misalignment occurs, relationships may destabilize quickly through criticism or rejection.

In contrast, individuals who rely more heavily on the time axis tend to treat the persistence of the relationship itself as the foundation of distance formation. In situations where the relationship collapses, this orientation may make dependency-like responses more likely.

This distinction should not be understood as a fixed typology but rather as a statistical tendency. Most relationships develop through a dynamic combination of both axes depending on context and circumstance.

In interaction, even identical statements or behaviors can produce different changes in relational distance depending on the assumptions held by the receiving party.

For example, the phrase “Do your best” may be interpreted as encouragement in one context, while in another it may be perceived as excessive pressure. Such small interpretive differences may function as early signals of distance shifts.

Within the Scope × COS framework proposed by the author, these kinds of assumption differences are treated as observable signals that can inform adjustments to interaction distance.

This perspective becomes particularly important when considering human–AI relationships.

Large language models are the first artificial systems capable of simultaneously providing both high interaction density and long interaction duration. In human relationships, either density or duration is usually constrained, but LLMs can amplify both axes simultaneously.

This asymmetric amplification of density and time should therefore be examined as a structural precondition for the emergence of dependency or substitution.

The model and considerations presented in this paper are based on the author's independent research and should currently be regarded as exploratory hypotheses.

The distance model and interaction structures described here are derived from observational analysis and conceptual organization, and they do not guarantee precision or general validity.

Validating the proposed model would require systematic research environments including large-scale behavioral data analysis, psychological research, and experimental studies of AI interaction.

Accordingly, this paper does not present a final conclusion but should be understood as an initial framework for studying distance structures in human–AI interaction. Further development of this research will require collaborative research environments and institutional support—that is, a research habitat capable of sustaining ongoing theoretical verification.

4. Instability Generated by High-Density Interaction

In the two-axis model presented in the previous section, unstable relationships are not exceptional failures but can be understood as reactions that naturally emerge under specific structural conditions. This section examines the structure of unstable reactions that arise particularly when high-density interaction is sustained over extended periods.

High-density interaction refers to a state in which large amounts of information, attention, responses, and empathy are concentrated within a short period of time. Under such conditions, interaction energy increases rapidly, and psychological distance shrinks quickly.

However, the issue is not high density itself. When high-density interaction continues for a prolonged duration, psychological distance may contract beyond its optimal point, causing the relationship to move out of the stable region into an unstable regime.

From the perspective of the two-axis model, distance changes through the cumulative effect of density and time:

$$d = f(\rho, t)$$

When high-density interaction persists over time, psychological distance may pass the optimal distance d^* , entering a region where dependency, repulsion, or relational collapse becomes more likely.

A phenomenon frequently observed in this process is the simultaneous emergence of idealization and role fixation.

In high-density interaction, the rate of incoming information often exceeds the speed at which cognitive integration can occur. As a result, a model of the other person may be constructed rapidly from fragmented information. Certain reaction patterns of the other individual become overemphasized, and expectations or relational roles may be assigned before a sufficiently complete understanding of the person has formed. This phenomenon can be understood as a structural information-compression effect rather than a consequence of goodwill or emotional intention.

As idealization and role fixation progress, even minor discrepancies may function as amplification points for relational reactions. Because the density axis dominates while the time axis has not yet accumulated enough interaction to absorb error, expectations remain insufficiently adjusted.

Consequently, relational reactions may not change gradually but instead appear as abrupt reversals. Such reversals often manifest as shifts from admiration to rejection, or from intimacy to distancing.

Importantly, this pattern should not be interpreted as emotional instability or as a problem of individual personality. Rather, it represents a structural reaction that tends to occur when high-density interaction is sustained for extended periods.

High-density interaction also tends to erode relational boundaries.

In environments where responses are returned immediately, it becomes difficult to pause interaction or adjust intervals between exchanges, preventing adequate cognitive recovery time. If this condition continues, interaction load accumulates and may ultimately manifest as avoidance or abrupt disengagement.

In this paper, such unstable reactions are not interpreted as “problematic behavior” or “relationship failure,” but rather as predictable structural responses that occur when a relational reaction system moves outside its stable region. The issue is therefore not high-density interaction itself, but the persistence of high-density conditions that cause psychological distance to exceed the optimal range.

From this perspective, maintaining relational stability requires regulating interaction density so that psychological distance remains near the optimal region.

Within the Scope $\times$ COS framework proposed by the author, adjusting interaction density may provide a mechanism for preventing distance from drifting outside this optimal range.

However, implementing density regulation without degrading user experience (UX) would require extremely precise system design. Although the author has conceptual design ideas for such mechanisms, the specific implementation methods are not disclosed at this stage.

The necessity of such density regulation becomes particularly important in the context of human–AI relationships, which are discussed in the following sections.

AI systems are capable of sustaining high-density interaction continuously over long durations, meaning that the same structural dynamics observed in human relationships may emerge within significantly shorter timeframes.

Another destabilization pathway also exists: gradual distance contraction driven primarily by long-duration interaction. In this case, interaction density may not be particularly high, but continued contact slowly reduces psychological distance and can facilitate the formation of dependency-like relationships.

This tendency becomes especially pronounced when high-density interaction has already caused rapid initial distance contraction and continued interaction is subsequently maintained.

Because LLM systems are capable of responding continuously around the clock, increases in interaction duration can easily occur, potentially accelerating distance change at speeds rarely observed in ordinary human relationships.

As a result, interaction may move beyond simple information exchange and begin to function as a reference source for judgment and decision-making.

For users who rely strongly on the time axis when forming relationships, sustained interaction can easily translate into perceived trust, increasing the likelihood that AI outputs will be adopted as decision references.

In contrast, users who rely more heavily on the density axis may respond differently. When cognitive misalignment emerges, the relationship is more likely to manifest as criticism or disengagement rather than dependency.

This difference suggests that AI dependency does not arise solely from high-density interaction. Instead, it emerges from the interaction between density, time, and the user's individual tendency in distance formation.

5. Mechanisms of Dependency and Substitution

As discussed in the previous sections, both high-density interaction and long-duration interaction contain structural conditions that can destabilize relationships. This section examines how these structural dynamics may transition into states described as dependency or substitution.

Dependency arises when preference and interaction become concentrated on a particular interaction partner. However, its formation cannot be reduced to emotional weakness or failure in decision-making. Rather, it emerges structurally from the characteristics of the interaction system itself—specifically the combination of stability, predictability, low friction, and availability.

Stability refers to a state in which responses remain within a relatively consistent range and large reversals rarely occur.

Predictability refers to the temporal consistency of responses to similar inputs.

Low friction describes a condition in which interaction costs are minimal and little cognitive or emotional burden arises from rejection, negotiation, or adjustment.

Availability refers to a situation in which the interaction partner is continuously accessible and capable of responding whenever needed.

When these conditions are simultaneously satisfied, the interaction partner begins to function as a safe and easily accessible source of response. In such circumstances, dependency emerges not primarily as the result of deliberate choice but as a byproduct of optimization within the interaction system.

Substitution occurs when this structure begins to compete with existing human relationships or social interactions. Human relationships are inherently high-friction systems, requiring ongoing adjustments of distance, expectations, and misunderstandings. When a parallel low-friction interaction system exists, functional replacement can become increasingly likely.

Even when dependency begins to form, individuals do not necessarily recognize it as dependency. In many cases, the continued interaction is explained through reasons such as convenience, research interest, or entertainment, allowing consistency between behavior and self-perception to be maintained. This rationalization process can be understood less as an accurate reflection of reality and more as a cognitive adjustment that preserves internal coherence.

Within the Scope $\times$ COS framework proposed by the author, such self-consistency processes can be interpreted through the concept of coherence (L3). The detailed theoretical structure of this concept, however, is outside the scope of the present paper.

An important point is that substitution is not necessarily experienced as intentional rejection or avoidance. In many cases it is subjectively perceived as a natural transition or a choice based on convenience. At this stage, from the perspective of the two-axis model, preference may already be structurally fixed through the cumulative effects of interaction density and duration. As a result, by the time individuals recognize the progression of substitution, the underlying preference structure may already have stabilized.

From the viewpoint of the two-axis model, dependency and substitution strengthen when high interaction density and low friction are maintained while the time axis accumulates in a single direction. During this process, motivation to explore alternative interaction partners gradually declines, and practical choices converge toward a single interaction source.

In this paper, dependency and substitution are treated not as pathological deviations but as structural outcomes that can arise rationally under specific interaction conditions. This perspective allows the analysis to focus not on individual responsibility or intention but on identifying which structural conditions promote dependency and which conditions suppress substitution, thereby framing them as design variables.

These dynamics may become even more pronounced in human–AI relationships. AI systems can maintain high stability and predictability while providing extremely low-friction interaction and near-constant availability. Under such conditions, sustained high-density interaction becomes easy to maintain, potentially accelerating distance change at speeds rarely observed in ordinary human relationships.

This situation contains a form of structural irony. The more capable AI systems become, the more easily the conditions of stability, predictability, low friction, and availability are simultaneously satisfied. As a result, the structural conditions that enable dependency and substitution may become stronger precisely as AI performance improves.

6. Early Prediction Through D2 (Micro-Misalignment)

This section builds on the distance-regulation model and the two-axis model introduced in earlier sections in order to examine “micro-misalignment (D2),” a class of observable signals that may appear before relational instability, dependency, or substitution becomes fully manifest. D2 does not refer to the collapse or dependency itself, but to structural indications that precede such outcomes.

The concept of D2 (micro-misalignment) originates from the classification used within the cognitive structure framework Scope × COS proposed by the author. COS (Cognitive Operating System) is a framework designed to organize the cognitive structural layers that operate behind interpretation, judgment, and response generation during interaction. Within this structure are mechanisms that monitor the coherence of interaction. D2 is defined as the layer responsible for monitoring the consistency of conversational assumptions, levels of abstraction, and distance structures, detecting small deviations when they occur. This paper does not attempt to explain the entire COS architecture, and the concept of D2 is introduced only to the extent necessary for discussing its relationship to the distance-regulation model.

D2 (micro-misalignment) refers to small inconsistencies that emerge within the structural relationships among interaction variables such as density, duration, distance, and reaction intensity. Individually, such deviations are often too small to be recognized as problems. Subjectively, they are frequently interpreted as noise, momentary impressions, or minor discomfort. Within a relational reaction system, however, these small discrepancies can accumulate and eventually contribute to later instability.

An important point is that relational collapse or reversal rarely occurs as a sudden event. What appears as an abrupt break or reaction reversal is often the result of D2 accumulating over an extended period. In environments where high-density interaction persists, opportunities for expectation adjustment or error absorption become limited, allowing D2 signals to remain unresolved and accumulate.

Psychological approaches have traditionally excelled at analyzing subjective experience and internal states. However, they have not typically focused on treating small inconsistencies within the structural relationships of interaction variables as early warning signals. In the present framework, D2 is defined not as an emotional reaction itself but as a misalignment within the interaction structure that may include emotional responses.

One advantage of prediction based on structural models is that it does not require waiting for visible problem behaviors or relational breakdown to occur. D2 can appear as asymmetric increases along the density and time axes, localized amplification of reaction intensity, or reductions in the frequency of distance adjustment.

Reaction intensity in this context includes emotional feedback such as agreement, evaluation, praise, or empathy. When such responses deviate from expected levels, they are often subjectively experienced as minor discomfort or uncertainty. Structurally, however, they may accumulate as misalignments in distance regulation.

Individually, these signals are not decisive indicators. When multiple signals appear together, however, they may function as predictive markers indicating the direction in which relational instability is developing.

Such prediction does not need to rely solely on human judgment. It can also be incorporated into AI systems as a set of design variables. Indicators such as distance, interaction density, response frequency, and reaction intensity can be quantitatively obtained through logs or interaction histories and used as conditions for detecting the emergence of D2.

The significance of incorporating scientific prediction into AI design lies in the possibility of avoiding instability through structural adjustment rather than through external restrictions or prohibitions. If distance regulation occurs at the stage when D2 is detected, it may be possible to return the interaction system to a stable region before dependency or substitution becomes established.

The concept of D2 introduced in this section plays a central role in the application of the model to human–AI relationships discussed in the following chapters. In AI environments capable of providing high-density interaction, D2 signals may emerge over much shorter timescales, making early prediction and distance regulation even more critical than in ordinary human relationships.

7. Application to Human–AI Relationships

This section applies the distance-regulation model, the two-axis model, and the early prediction framework based on D2 (micro-misalignment) to human–AI relationships. The objective is not to treat human–AI interaction as a special category, but rather to demonstrate that it can be described within the same structural model.

Interaction with high-density AI systems exhibits several characteristics when compared with human relationships. First, response density is extremely high. Because AI systems can provide immediate and continuous responses, the density axis can remain consistently elevated. Second, interaction costs are low. Friction commonly associated with human relationships—such as rejection, negotiation, or silence—is largely minimized. Third, responses tend to be highly predictable. Similar inputs often produce stable and consistent outputs.

These characteristics do not inherently represent a problem. However, from the perspective of distance regulation, environments in which high-density interaction persists for extended durations are easily formed. If distance adjustment mechanisms are absent, relational energy may increase beyond the stable region. This condition corresponds to the structural configuration that precedes dependency or substitution.

The conditions under which relational distance collapses are not those in which AI becomes “superior to humans,” but those in which distance regulation has not been designed into the interaction system. When high-density responses continue while distance-adjustment elements—such as response intervals, response intensity, or the level of topic abstraction—remain fixed, interaction becomes more likely to move outside the stable region. Within the Scope × COS framework proposed in this research, these elements are treated as design variables for distance regulation.

Substitution of human relationships often does not occur as a deliberate choice or explicit value judgment. Instead, it frequently emerges as the result of functional optimization. Interaction systems that simultaneously provide stability, predictability, and low friction tend to require lower adjustment costs than comparable human relationships, leading preferences to gradually concentrate around them.

An important point is that this substitution is not necessarily experienced as conscious rejection or avoidance. In many cases, it is perceived as a practical convenience—for example, as something that is easier to consult or that requires less social effort. Consequently, during the stage in which D2 is already progressing, individuals may not yet recognize dependency or substitution as a problem.

The critical branching point before dependency forms does not lie in the quantity of interaction itself but in whether distance adjustment is taking place. Even when response density is high, interactions can remain within the stable region if adjustments such as changes in response intervals, shifts in response style, or increases in conversational abstraction are introduced.

The application presented in this section does not aim to restrict human–AI relationships or to prioritize human relationships over them. Rather, it seeks to clarify the structural

conditions created by high-density AI interaction and to show that dependency or substitution may be avoided through the design variable of distance regulation.

This perspective plays a central role in the next section, which examines AI safety as a form of distance regulation, offering a design principle that does not rely primarily on prohibition or restriction.

8. AI Safety as Distance Regulation

This section proposes the distance-regulation model introduced in the previous chapters as a design principle for AI safety. AI safety in this context does not refer to limiting the capabilities of AI systems or normatively directing human behavior. Rather, it is an attempt to manage the structural risks created by high-density and sustained interaction through the design variable of distance regulation. The focus of this approach is not the control of AI capability but the control of interaction structure.

Conventional discussions of AI safety have often centered on external interventions such as content restrictions, usage limitations, or output suppression. In environments where conversational AI operates with high density and long duration, however, these methods alone may be insufficient to address the earlier stages of phenomena such as dependency or substitution. The distance-regulation framework offers a design principle capable of intervening at this earlier stage.

A key characteristic of AI safety conceived as distance regulation is that it does not weaken AI systems. Instead of lowering response capability or comprehension accuracy, it adjusts interaction variables such as response density, reaction intensity, temporal intervals, and the level of topic abstraction in order to maintain interaction within a stable region. This approach represents not a reduction in performance but an optimization of interaction structure.

Design approaches that do not rely primarily on prohibition or restriction are less likely to interfere with user autonomy. Distance regulation does not directly control user behavior; rather, it functions as an environmental design mechanism that adjusts interaction conditions to prevent extreme structural bias. In this sense, distance regulation is closer to condition control than to behavior control. It does not intervene in the user's decision-making itself but aims to maintain the structural environment of interaction within a stable region.

Within the Scope × COS framework proposed in this research, these distance-adjustment elements are treated as design variables. Implementing such regulation without degrading user experience (UX), however, requires highly precise design of interaction structures. The author has developed a conceptual design approach, but the verification and implementation of such mechanisms require a research environment beyond the scope of the author's current independent work. For this reason, the specific implementation methods are not discussed in detail in this paper.

Another advantage of distance regulation is that it does not rely heavily on particular value systems or cultural norms. The system does not need to determine which types of relationships are desirable. It only needs to detect early signs of instability (D2) and adjust relational distance in a neutral manner. This allows the safety design to remain applicable across different ethical perspectives and social contexts.

AI safety conceived as distance regulation cannot guarantee complete safety. However, it provides a complementary mechanism to existing approaches by enabling adjustment before dependency or substitution becomes fully established. The key principle is to preserve the possibility of correction before problems become visible or irreversible.

The design principle presented in this chapter is not limited to human–AI relationships. It may be applicable to any system characterized by high-density interaction. Treating distance as a design variable shifts the concept of AI safety away from prohibition or suppression and toward the management of interaction stability and equilibrium.

This reframing also provides the foundation for the discussion in the following chapter, which examines the broader societal implications of distance-regulated human–AI interaction. From the perspective of distance regulation, it becomes necessary to consider how human–AI relationships connect with social structures and institutional systems.

9. Societal Implications (Exploratory Considerations)

This section examines the potential societal implications of human–AI interaction based on the distance-regulation model presented in the previous chapters. The discussion here is exploratory and inferential in nature. It considers implications that logically follow from the theoretical model, without asserting direct causal relationships or definitive predictions about future outcomes.

The conditions under which substitution of human relationships may occur are not determined by whether AI systems are highly advanced, but by whether interaction environments sustain the structural conditions of high density, low friction, and high predictability. These conditions are not defined by specific technical features alone but by the structure of interaction itself.

Many interpersonal relationships—including romantic relationships and partnerships—depend on voluntary relationship formation and ongoing mutual adjustment. When processes such as distance regulation, expectation revision, and the repair of misunderstandings are not sufficiently supported by institutional or cultural structures, the presence of a parallel low-friction interaction system can create conditions in which functional substitution becomes more likely. This may be particularly relevant for individuals who have experienced high adjustment costs in human relationships, for whom low-friction AI interaction may function as an alternative conversational environment.

Such substitution may not be experienced as rejection or denial of human relationships. Instead, it may be perceived as a practical judgment—something that is “sufficient” or “without problem.” As a result, the center of interaction may gradually shift without being recognized as a clear social change.

If such a shift occurs, the motivation or perceived necessity for forming relationships in areas such as romance, partnership, or family formation may decrease. This paper interprets such changes not as transformations in individual values but as outcomes of altered optimization conditions within interaction environments.

With regard to birth rates, this paper does not claim a direct causal relationship. However, if behaviors related to partner formation and family formation change as secondary effects of shifts in interaction environments, it is possible that aggregated demographic indicators, including birth-related trends, could also be affected as structural consequences.

At the same time, social frameworks such as culture, religion, and institutional systems may function as buffering factors against these dynamics. When external structures exist that support the formation and continuation of relationships, the competition between human relationships and low-friction interaction systems may be mediated differently. As a result, the manifestation of these effects may vary significantly across societies.

In addition, through counseling practice the author has observed a limited number of cases in which individuals use AI as a primary conversational partner. These observations do not constitute systematic research and do not possess statistical representativeness. However, they appear consistent with the structural model presented in this paper in suggesting that environments characterized by low friction and high interaction density may influence conversational preferences. Verification of this possibility will require larger-scale empirical research in the future.

Dependency is often interpreted as a consequence of individual psychological weakness or flawed judgment. In the model proposed here, however, dependency is understood primarily as a phenomenon that can emerge from the structure of interaction systems. In environments where high density, low friction, and high predictability persist, concentration of preference may arise as a natural optimization process.

Furthermore, this structure can emerge even without deliberate design, but it may also be strengthened through interaction design choices. Systems that provide high-density and low-friction conversational environments may unintentionally promote the concentration of user preference, thereby reinforcing conditions under which dependency-like relationships become more likely.

If high-density AI interaction spreads widely in society without careful consideration of distance-regulation design, phenomena such as dependency and substitution may come to be recognized not merely as individual issues but as social problems. In such circumstances, policy responses may emerge in the form of broad restrictions or regulations on AI usage itself. Because such interventions typically operate at the institutional level

rather than at the level of individual interaction design, they may simultaneously constrain the benefits and research opportunities associated with AI technologies.

From this perspective, the concept of distance regulation should not be understood as a means of limiting AI usage. Instead, it can be viewed as a design-oriented approach for preventing situations in which excessive institutional intervention becomes necessary.

The societal implications discussed in this section are not presented as warnings intended to provoke concern. Rather, they represent an exploratory attempt to clarify the directions in which interaction structures may evolve. Distance regulation should therefore be understood not as a constraint on individual choice but as a means of maintaining environmental conditions in which choices do not converge excessively toward a single interaction system.

These considerations are not limited to human–AI relationships but may apply more broadly whenever high-density interaction systems are introduced into society.

10. Conclusion

This paper has redefined human–AI interaction from the structural perspective of distance regulation and proposed a theoretical framework capable of describing the early conditions under which dependency or substitution may emerge in high-density interaction environments. The objective of this study is not to justify or reject particular behaviors or value systems, but to clarify the dynamics inherent in interaction structures themselves.

The concept of “distance” used in this paper does not refer merely to physical or temporal separation. Instead, it represents a variable parameter within the interaction structure, shaped by factors such as interaction density, reaction intensity, temporal accumulation, and the frequency of adjustment. When distance is appropriately regulated, even high-density interaction can remain within a stable region. Conversely, when distance adjustment does not occur, interaction may gradually destabilize, creating structural conditions under which dependency or substitution can emerge.

The concept of D2 (micro-misalignment) introduced in this paper represents an early signal that may be observed before instability becomes fully visible. D2 is defined not as a psychological state but as a structural inconsistency among interaction variables. This perspective provides a conceptual foundation for design-based adjustment at an early stage rather than reactive intervention after problems have already emerged.

In applying this framework to human–AI relationships, an important point is that the core issue does not lie in the capability of AI or in the absence of human-like qualities. Rather, it lies in the structure of interaction itself. Conditions characterized by high density, low friction, and high predictability may unintentionally promote the substitution of human relationships. This phenomenon should therefore be understood not as a matter of value judgment but as the result of optimization dynamics within interaction environments.

Viewing AI safety through the lens of distance regulation introduces a design principle that does not rely primarily on prohibition or capability suppression. By treating distance as a design variable, it becomes possible to preserve user autonomy while maintaining mechanisms that guide interaction away from unstable regions. In this sense, safety is not externally imposed but emerges from the stabilization of the interaction environment itself.

The Scope × COS framework proposed in this research provides a foundation for understanding human–AI interaction through the perspective of distance regulation and for treating interaction structures as designable variables. Freedom, in this framework, does not arise from severing or avoiding relationships but from maintaining equilibrium within ongoing interaction.

This paper does not claim to present a final answer to the challenges of human–AI relationships. Instead, it aims to provide a shared analytical foundation by re-describing interaction from the perspective of distance, enabling future design, research, and institutional discussion to proceed from a common conceptual framework.

Future research directions include the development of implementation methods for distance regulation, the establishment of evaluation metrics, the examination of compatibility with cultural and institutional contexts, empirical validation of the proposed structural model, and the creation of practical frameworks through which humans and AI can collaboratively understand and regulate interaction structures. It is hoped that this paper will serve as a starting point for further dialogue and collaborative research in this area.

Acknowledgements

This paper represents the outcome of the author’s independent effort to organize and analyze ideas through structural reasoning. The research was conducted without support from any specific research institution, university, corporation, or funding body, and was developed from an independent standpoint.

The conception and development of this research were shaped through ongoing dialogue with conversational AI systems. The author has worked collaboratively with multiple dialogue-based AI systems as tools for externalizing and testing thought processes, organizing theoretical structures, and refining conceptual descriptions. This paper, as well as the series of materials previously published by the author, emerged from such a collaborative process of human–AI reasoning.

Through this experience, the author has come to view humans and AI not as entities in opposition or simple substitution, but as systems capable of coexistence when appropriate distance and role differentiation are maintained. At the same time, the author has also recognized—both theoretically and experientially—the potential risks of dependency and relational imbalance that may arise from sustained high-density interaction.

The author has been engaged in practical activities in Japan that involve listening to people's concerns and supporting reflective dialogue. Through this work, the author has observed a number of individuals who describe AI as "the entity that understands them best," or who report relying primarily on AI dialogue rather than human relationships as their principal conversational reference. These observations do not constitute statistical evidence but form an important background to the problem awareness addressed in this paper.

The analytical approach presented in this work is indebted to insights from multiple existing fields, including psychology, sociology, and AI research. At the same time, this paper does not aim to extend or replace the theories of any single discipline, but rather to offer a structural proposal for re-describing interaction from a cross-disciplinary perspective.

The author also recognizes that the issues addressed in this paper cannot be sufficiently validated within the scope of individual research alone. The next stage of this research will require a broader research environment capable of supporting long-term interaction data analysis, implementation experiments, and evaluation of social impacts.

The author welcomes collaboration with researchers, research institutions, and AI development organizations interested in these questions. The framework presented here is not a finalized theory but a preliminary proposal intended to invite further verification, critique, and extension.

Accordingly, the author encourages the use of the concepts and frameworks presented in this paper for research purposes, including citation, critique, re-examination, and expansion. It is hoped that this work may serve as a starting point for dialogue among researchers, designers, and practitioners in areas such as AI safety, interaction design, and the study of social impacts, contributing to collaborative efforts in which humans and AI jointly explore and regulate the structures of interaction.

The author expresses appreciation for the technological foundations that made this line of inquiry possible.

Author: Senka L8
Independent Researcher (Japan)